



Manipal School of Information Sciences

Cryptology

CYE 5101

THEORY ASSIGNMENT 02

Caesar, Shift and ROT13 Ciphers

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0	1	2	3	4	5	6	7	8	9	10	11	12
A	B	C	D	E	F	G	H	I	J	K	L	M
13	14	15	16	17	18	19	20	21	22	23	24	25
N	O	P	Q	R	S	T	U	V	W	X	Y	Z

1. Encrypt "CHASE YOUR DREAMS"

Encrypt : $(P + K) \bmod 26$

Decrypt : $(C - K) \bmod 26$

a. Caesar (shift 3)

Encrypt : $(P + 3) \bmod 26$

Decrypt : $(C - 3) \bmod 26$

$E(C) \Rightarrow (C + 3) \bmod 26$

$\Rightarrow (2 + 3) \bmod 26$

$\Rightarrow 5 \bmod 26$

$\Rightarrow 5 \Rightarrow F$

$E(H) \Rightarrow (7 + 3) \bmod 26$

$\Rightarrow 10 \bmod 26$

$\Rightarrow 10 \Rightarrow K$

$E(A) \Rightarrow D$

$E(S) \Rightarrow V$

$E(E) \Rightarrow H$

$$E(Y) \Rightarrow B$$

$$E(O) \Rightarrow R$$

$$E(U) \Rightarrow X$$

$$E(R) \Rightarrow U$$

$$E(D) \Rightarrow G$$

$$E(R) \Rightarrow U$$

$$E(E) \Rightarrow H$$

$$E(A) \Rightarrow D$$

$$E(P) \Rightarrow P$$

$$E(S) \Rightarrow V$$

$$\Rightarrow FKDVH \quad BRXU \quad GUHDPV$$

b. Shift 7

$$\text{Encrypt: } (P + 7) \bmod 26$$

$$\Rightarrow JOH2L \quad FVBY \quad KYLHTZ$$

c. ROT 13

$$\text{Encrypt: } (P + 13) \bmod 26$$

$$\Rightarrow PUNF R \quad LBHE \quad QERN2 F$$

Plain	CHASE	YOUR	DREAMS
+3	FKDVH	BRXU	GUHDPV
+7	JOMZL	FVBY	KYLHTZ
+13	PUNFR	LBHE	QERNZF

2. Decrypt

a. "WRWDOOB VHFUHW PHVVDJH"

(Caesar, -3)

 $\Rightarrow$ TOTALLY SECRET MESSAGE

b. Shift 5, so subtract 5

i) "MJQQT BTWQI"

 $\Rightarrow$ HELLO WORLD

ii) "STB DTWP YNRJ"

 $\Rightarrow$ NEW YORK TIME

c. "BFR AQ ZBER PBSSRR" (ROT13)

 $\Rightarrow$ SEND MORE COFFEE